Министерство образования Российской Федерации
Пензенский государственный университет
Кафедра „Вычислительная техника“

ОТЧЕТ

по лабораторной работе №5

по курсу „Разработка кроссплатформенных приложений “

на тему „ Мнопоточность в Java“

Вариант 7

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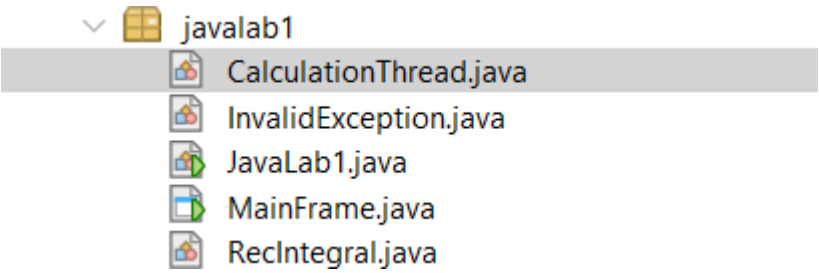
Цель работы: научиться создавать многопоточные приложения с использованием стандартных средств языка Java.

Задание: Модифицировать приложение из предыдущей лабораторной работы, реализовав вычисление определенного интеграла в нескольких дополнительных потоках (число потоков определяется номером варианта), снимая нагрузку с основного потока и предотвращая "подвисание" графического интерфейса. Варианты с номерами до 5 включительно реализуют многопоточность путем наследования от класса Thread, остальные реализуют интерфейс Runnable. Оформление лабораторной работы должно быть выполнено в соответствии с требованиями, приведенными в Приложении 2.

Ход работы:



Фигура 1: Создали новый класс CalculationThread

Изменили функцию ClacActionPerformed:

```
package javalab1;

import java.io.*;
import java.util.ArrayList;
import javax.swing.JFileChooser;
import javax.swing.JOptionPane;
import javax.swing.table.DefaultTableModel;


public class MainFrame extends javax.swing.JFrame {

    // Коллекция для хранения записей
    private final ArrayList<RecIntegral> dataList = new ArrayList<>();

    /**
     * Creates new form MainFrame
     */
    public MainFrame() {
```

```
    initComponents();
    setTitle("Лабораторная работа");
    setLocationRelativeTo(null);
}

/**
 * This method is called from within the constructor to initialize the form.
 * WARNING: Do NOT modify this code. The content of this method is always
 * regenerated by the Form Editor.
 */
@SuppressWarnings("unchecked")
// <editor-fold defaultstate="collapsed" desc="Generated Code">
private void initComponents() {

    jPanel1 = new javax.swing.JPanel();
    uppr = new javax.swing.JTextField();
    downpr = new javax.swing.JTextField();
    st3p = new javax.swing.JTextField();
    jLabel1 = new javax.swing.JLabel();
    jLabel2 = new javax.swing.JLabel();
    jLabel3 = new javax.swing.JLabel();
    jPanel2 = new javax.swing.JPanel();
    jPanel4 = new javax.swing.JPanel();
    jLabel4 = new javax.swing.JLabel();
    add = new javax.swing.JButton();
    delete = new javax.swing.JButton();
    clac = new javax.swing.JButton();
    clearBtn = new javax.swing.JButton();
    fillBtn = new javax.swing.JButton();
    saveTxtBtn = new javax.swing.JButton();
    loadTxtBtn = new javax.swing.JButton();
    saveBinBtn = new javax.swing.JButton();
    loadBinBtn = new javax.swing.JButton();
    jPanel3 = new javax.swing.JPanel();
    jScrollPane4 = new javax.swing.JScrollPane();
    jTable1 = new javax.swing.JTable();

    setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
    setTitle("Лабораторная работа - f(x) = 1/ln(x)");

    jPanel1.setBackground(new java.awt.Color(153, 204, 255));

    uppr.setColumns(10);

    downpr.setColumns(10);
```

```
        st3p.setColumns(10);

        jLabel1.setText("Верх. пр.");

        jLabel2.setText("Ниж. пр.");

        jLabel3.setText("Шаг");

        javax.swing.GroupLayout jPanel1Layout = new
javax.swing.GroupLayout(jPanel1);
        jPanel1.setLayout(jPanel1Layout);
        jPanel1Layout.setHorizontalGroup(

jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(javax.swing.GroupLayout.Alignment.TRAILING,
jPanel1Layout.createSequentialGroup()
                .addContainerGap(20, Short.MAX_VALUE)
                .addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.A
lignment.LEADING)
                    .addComponent(jLabel1,
javax.swing.GroupLayout.Alignment.TRAILING)
                    .addComponent(jLabel2,
javax.swing.GroupLayout.Alignment.TRAILING)
                    .addComponent(jLabel3,
javax.swing.GroupLayout.Alignment.TRAILING))
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATE
D)
                .addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.A
lignment.LEADING)
                    .addComponent(uppr, javax.swing.GroupLayout.PREFERRED_SIZE,
88, javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addComponent(downpr,
javax.swing.GroupLayout.PREFERRED_SIZE, 88,
javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addComponent(st3p, javax.swing.GroupLayout.PREFERRED_SIZE,
88, javax.swing.GroupLayout.PREFERRED_SIZE))
                .addGap(40, 40, 40))
        );
        jPanel1Layout.setVerticalGroup(

jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(jPanel1Layout.createSequentialGroup()
                .addGap(20, 20, 20)
```

```
            .addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.A
lignment.BASELINE)
                .addComponent(uppr, javax.swing.GroupLayout.PREFERRED_SIZE,
25, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addComponent(jLabel1))
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATE
D)
            .addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.A
lignment.BASELINE)
                .addComponent(downpr,
javax.swing.GroupLayout.PREFERRED_SIZE, 25,
javax.swing.GroupLayout.PREFERRED_SIZE)
                .addComponent(jLabel2))
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATE
D)
            .addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.A
lignment.BASELINE)
                .addComponent(st3p, javax.swing.GroupLayout.PREFERRED_SIZE,
25, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addComponent(jLabel3))
            .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE))
        );

        jPanel2.setBackground(new java.awt.Color(153, 204, 255));

        jPanel4.setBackground(new java.awt.Color(153, 153, 255));

        jLabel4.setText("Функция: 1/ln(x)");

        javax.swing.GroupLayout jPanel4Layout = new
javax.swing.GroupLayout(jPanel4);
        jPanel4.setLayout(jPanel4Layout);
        jPanel4Layout.setHorizontalGroup(

jPanel4Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(jPanel4Layout.createSequentialGroup()
                .addGap(40, 40, 40)
                .addComponent(jLabel4)
                .addContainerGap(40, Short.MAX_VALUE))
        );
        jPanel4Layout.setVerticalGroup(

jPanel4Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(jPanel4Layout.createSequentialGroup()
```

```
            .addGap(20, 20, 20)
            .addComponent(jLabel4, javax.swing.GroupLayout.PREFERRED_SIZE,
49, javax.swing.GroupLayout.PREFERRED_SIZE)
            .addContainerGap(20, Short.MAX_VALUE))
        );

        add.setText("Добавить");
        add.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                addActionPerformed(evt);
            }
        });

        delete.setText("Удалить");
        delete.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                deleteActionPerformed(evt);
            }
        });

        clac.setText("Вычислить");
        clac.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                clacActionPerformed(evt);
            }
        });

        clearBtn.setText("Очистить");
        clearBtn.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                clearActionPerformed(evt);
            }
        });

        fillBtn.setText("Заполнить");
        fillBtn.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                fillActionPerformed(evt);
            }
        });

        saveTxtBtn.setText("Сохранить(текст)");
        saveTxtBtn.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                saveTxtBtnActionPerformed(evt);
```

```
        }
    });

    loadTxtBtn.setText("Загрузить(текст)");
    loadTxtBtn.addActionListener(new java.awt.event.ActionListener() {
        public void actionPerformed(java.awt.event.ActionEvent evt) {
            loadTxtBtnActionPerformed(evt);
        }
    });

    saveBinBtn.setText("Сохранить(Двоичный)");
    saveBinBtn.addActionListener(new java.awt.event.ActionListener() {
        public void actionPerformed(java.awt.event.ActionEvent evt) {
            saveBinBtnActionPerformed(evt);
        }
    });

    loadBinBtn.setText("Загрузить(Двоичный)");
    loadBinBtn.addActionListener(new java.awt.event.ActionListener() {
        public void actionPerformed(java.awt.event.ActionEvent evt) {
            loadBinBtnActionPerformed(evt);
        }
    });

    javax.swing.GroupLayout jPanel2Layout = new
javax.swing.GroupLayout(jPanel2);
    jPanel2.setLayout(jPanel2Layout);
    jPanel2Layout.setHorizontalGroup(

jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(jPanel2Layout.createSequentialGroup()
            .addComponent(jPanel4, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
            .addGap(24, 24, 24)
            .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.A
lignment.LEADING, false)
                .addComponent(fillBtn, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                .addComponent(clearBtn, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                .addComponent(clac, javax.swing.GroupLayout.DEFAULT_SIZE, 130,
Short.MAX_VALUE)
                .addComponent(delete, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
```

```
                .addComponent(add, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
            .addGap(18, 18, 18)
            .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.A
lignment.LEADING, false)
                .addComponent(saveTxtBtn,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                .addComponent(loadTxtBtn,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                .addComponent(saveBinBtn,
javax.swing.GroupLayout.Alignment.TRAILING,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                .addComponent(loadBinBtn,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
            .addContainerGap())
        );
        jPanel2Layout.setVerticalGroup(

jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(jPanel2Layout.createSequentialGroup()
            .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.A
lignment.LEADING)
                .addGroup(jPanel2Layout.createSequentialGroup()
                    .addGap(10, 10, 10)
                    .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLay
out.Alignment.BASELINE)
                        .addComponent(add)
                        .addComponent(saveTxtBtn))
                    .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RE
LATED)
                    .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLay
out.Alignment.LEADING, false)
                        .addComponent(delete,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                        .addComponent(loadTxtBtn,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
                    .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RE
LATED)
```

```
                    .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLay
out.Alignment.BASELINE)
                        .addComponent(clac)
                        .addComponent(saveBinBtn))
                    .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RE
LATED)
                    .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLay
out.Alignment.BASELINE)
                        .addComponent(clearBtn)
                        .addComponent(loadBinBtn)))
                .addGroup(jPanel2Layout.createSequentialGroup()
                    .addContainerGap()
                    .addComponent(jPanel4,
javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)))
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATE
D)
            .addComponent(fillBtn)
            .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE))
        );

        jTable1.setAutoCreateRowSorter(true);
        jTable1.setBackground(new java.awt.Color(204, 204, 255));
        jTable1.setModel(new javax.swing.table.DefaultTableModel(
            new Object [][] {

            },
            new String [] {
                "Нижний предел", "Верхний предел", "Шаг", "Результат"
            }
        ) {
            boolean[] canEdit = new boolean [] {
                true, true, true, false
            };

            public boolean isCellEditable(int rowIndex, int columnIndex) {
                return canEdit [columnIndex];
            }
        });
        jScrollPane4.setViewportView(jTable1);

        javax.swing.GroupLayout jPanel3Layout = new
javax.swing.GroupLayout(jPanel3);
```

```
        jPanel3.setLayout(jPanel3Layout);
        jPanel3Layout.setHorizontalGroup(

jPanel3Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addComponent(jScrollPane4)
        );
        jPanel3Layout.setVerticalGroup(

jPanel3Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(javax.swing.GroupLayout.Alignment.TRAILING,
jPanel3Layout.createSequentialGroup()
                .addComponent(jScrollPane4,
javax.swing.GroupLayout.PREFERRED_SIZE, 275,
javax.swing.GroupLayout.PREFERRED_SIZE)
                .addGap(0, 0, Short.MAX_VALUE))
        );

        javax.swing.GroupLayout layout = new
javax.swing.GroupLayout(getContentPane());
        getContentPane().setLayout(layout);
        layout.setHorizontalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(layout.createSequentialGroup()
                .addComponent(jPanel1, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATE
D, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                .addComponent(jPanel2, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
            .addComponent(jPanel3, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
        );
        layout.setVerticalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(layout.createSequentialGroup()
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignmen
t.TRAILING, false)
                    .addComponent(jPanel2, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                    .addComponent(jPanel1, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATE
D)
```

```
            .addComponent(jPanel3, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
        );

        pack();
        setLocationRelativeTo(null);
    }// </editor-fold>

    private void addActionPerformed(java.awt.event.ActionEvent evt) {
        try {
            double low = Double.parseDouble(downpr.getText());
            double up = Double.parseDouble(uppr.getText());
            double stepVal = Double.parseDouble(st3p.getText());

            RecIntegral rec = new RecIntegral(low, up, stepVal);
            dataList.add(rec);

            DefaultTableModel    tModel = (DefaultTableModel) jTable1.getModel();
            tModel.addRow(new Object[]{low, up, stepVal, ""});

        }
      catch (InvalidException ex) {
    JOptionPane.showMessageDialog(
        this,
        ex.getMessage() + "\nНекорректное значение: " + ex.getErrVal(),
        "Ошибка диапазона",
        JOptionPane.ERROR_MESSAGE
    );
}

        catch (NumberFormatException e) {
            JOptionPane.showMessageDialog(this, "Введите корректные числовые
значения!", "Ошибка ввода", JOptionPane.ERROR_MESSAGE);

        }
    }

    private void deleteActionPerformed(java.awt.event.ActionEvent evt) {
        int selectedRow = jTable1.getSelectedRow();
        if (selectedRow != -1) {
            dataList.remove(selectedRow);
            ((DefaultTableModel) jTable1.getModel()).removeRow(selectedRow);
        } else {
            JOptionPane.showMessageDialog(this, "Выберите строку для удаления!");
```

```
        }
    }

    private void clacActionPerformed(java.awt.event.ActionEvent evt) {
DefaultTableModel tModel = (DefaultTableModel) jTable1.getModel();

    dataList.clear();

    for (int i = 0; i < tModel.getRowCount(); i++) {
        Object lowObj = tModel.getValueAt(i, 0);
        Object upObj = tModel.getValueAt(i, 1);
        Object stepObj = tModel.getValueAt(i, 2);

        if (lowObj == null || upObj == null || stepObj == null) continue;

        try {
            double low = Double.parseDouble(lowObj.toString());
            double up = Double.parseDouble(upObj.toString());
            double stepVal = Double.parseDouble(stepObj.toString());

            // Создаем объект записи
            RecIntegral rec = new RecIntegral(low, up, stepVal);

            dataList.add(rec);

            CalculationThread task = new CalculationThread(rec, tModel, i);
            Thread thread = new Thread(task);
            thread.start();

        } catch (NumberFormatException e) {
            JOptionPane.showMessageDialog(this, "Некорректный ввод числа в строке
" + (i + 1), "Ошибка", JOptionPane.ERROR_MESSAGE);
        } catch (InvalidException exe) {
            JOptionPane.showMessageDialog(this, exe.getMessage(), "Ошибка
диапазона", JOptionPane.ERROR_MESSAGE);
        }
    }

    }

    private void clearActionPerformed(java.awt.event.ActionEvent evt) {
        ((DefaultTableModel) jTable1.getModel()).setRowCount(0);
    }

    private void fillActionPerformed(java.awt.event.ActionEvent evt) {
```

```
        DefaultTableModel tModel = (DefaultTableModel) jTable1.getModel();
        tModel.setRowCount(0);
        for (RecIntegral rec : dataList) {
            tModel.addRow(new Object[]{
                rec.getLowerBound(),
                rec.getUpperBound(),
                rec.getStep(),
                rec.getResult()
            });
        }
    }

    private void saveTxtBtnActionPerformed(java.awt.event.ActionEvent evt) {
        javax.swing.JFileChooser fc = new javax.swing.JFileChooser();
        fc.setCurrentDirectory(new java.io.File("C:\\Users\\alex\\Desktop\\for
university\\3 kyrs\\6 семетр\\java\\тесты"));
        if (fc.showSaveDialog(this) == javax.swing.JFileChooser.APPROVE_OPTION)
{
            java.io.File file = fc.getSelectedFile();

            if (!file.getName().toLowerCase().endsWith(".txt")) {
                file = new java.io.File(file.getAbsolutePath() + ".txt");
            }

            try (java.io.FileWriter fw = new java.io.FileWriter(file)) {
                for (RecIntegral rec : dataList) {
                    fw.write(rec.toString() + "\n");
                }
            } catch (java.io.IOException ex) {
                javax.swing.JOptionPane.showMessageDialog(
                    this,
                    "Ошибка записи: " + ex.getMessage()
                );
            }
        }

    }

    private void loadTxtBtnActionPerformed(java.awt.event.ActionEvent evt) {
        javax.swing.JFileChooser fc = new javax.swing.JFileChooser();
         fc.setCurrentDirectory(new java.io.File("C:\\Users\\alex\\Desktop\\for
university\\3 kyrs\\6 семетр\\java\\тесты"));
        if (fc.showOpenDialog(this) ==
javax.swing.JFileChooser.APPROVE_OPTION) {
```

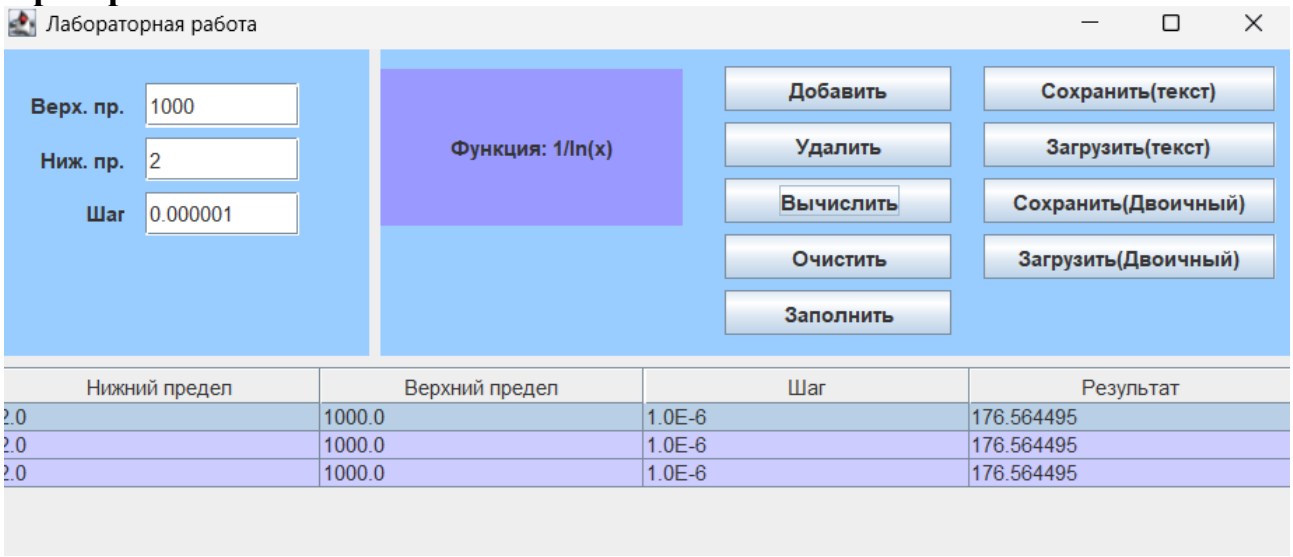
```
            try (java.io.BufferedReader br = new java.io.BufferedReader(new
java.io.FileReader(fc.getSelectedFile()))) {
                dataList.clear();
                String line;
                while ((line = br.readLine()) != null) {
                    String[] p = line.split(" ");
                    if (p.length >= 3) {
                        RecIntegral rec = new RecIntegral(
                            Double.parseDouble(p[0]),
                            Double.parseDouble(p[1]),
                            Double.parseDouble(p[2])
                        );
                        if (p.length == 4) rec.setResult(p[3]);
                        dataList.add(rec);
                    }
                }
                fillActionPerformed(evt);
            } catch (Exception ex) {
                javax.swing.JOptionPane.showMessageDialog(this, "Ошибка чтения: " +
ex.getMessage());
            }
        }

        // TODO add your handling code here:
    }

    private void saveBinBtnActionPerformed(java.awt.event.ActionEvent evt) {
    JFileChooser fc = new JFileChooser();
    fc.setCurrentDirectory(new java.io.File("C:\\Users\\alex\\Desktop\\for
university\\3 kyrs\\6 семетр\\java\\тесты"));
    if (fc.showSaveDialog(this) == JFileChooser.APPROVE_OPTION) {
        java.io.File file = fc.getSelectedFile();

        if (!file.getName().toLowerCase().endsWith(".bin")) {
            file = new java.io.File(file.getAbsolutePath() + ".bin");
        }

        try (ObjectOutputStream oos = new ObjectOutputStream(new
FileOutputStream(file))) {
            oos.writeObject(dataList);
        } catch (IOException ex) {
            JOptionPane.showMessageDialog(
                this,
                "Ошибка сериализации: " + ex.getMessage()
            );
```

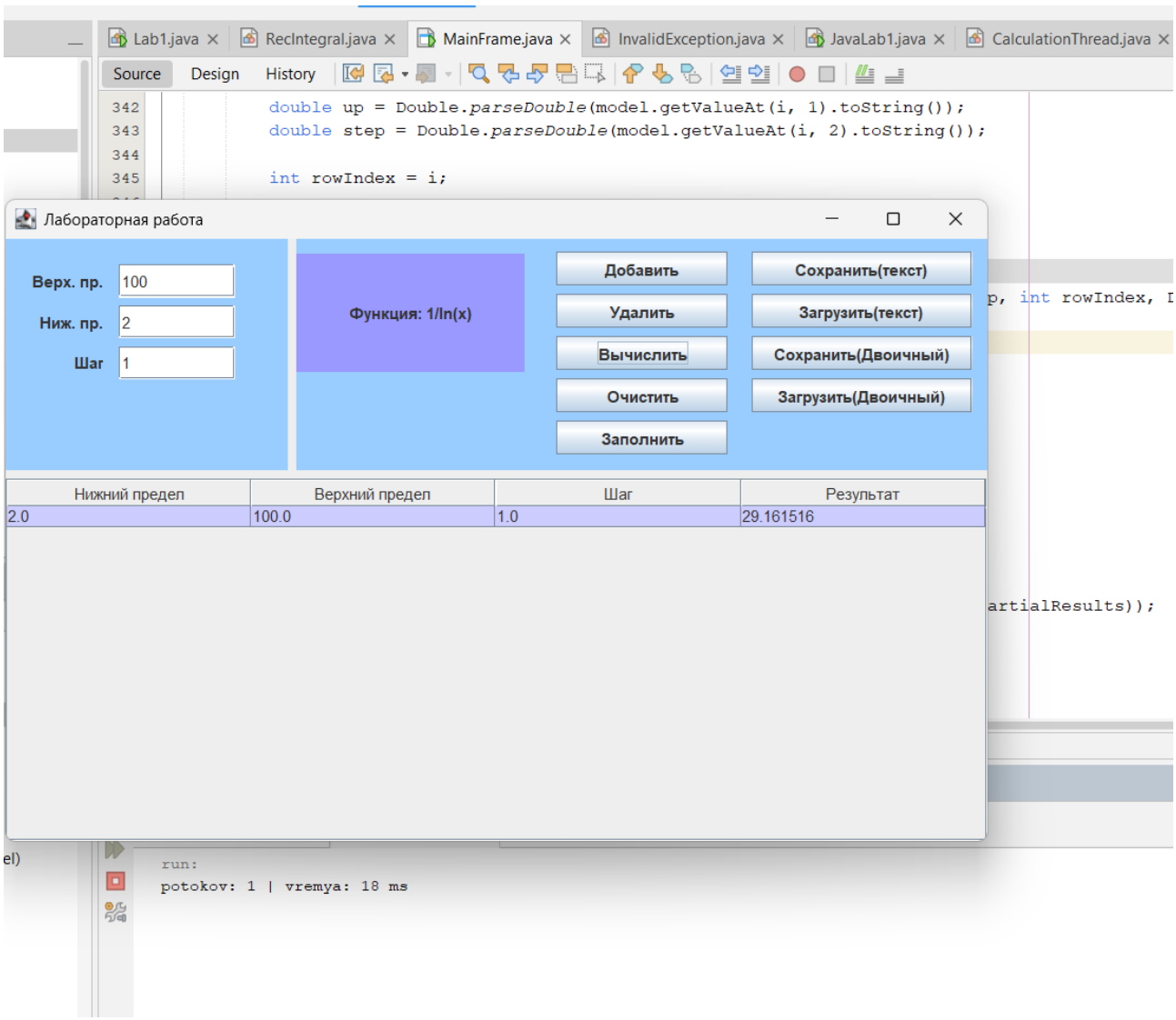
```
        }
    }

    }

    private void loadBinBtnActionPerformed(java.awt.event.ActionEvent evt) {
javax.swing.JFileChooser fc = new javax.swing.JFileChooser();
fc.setCurrentDirectory(new java.io.File("C:\\Users\\alex\\Desktop\\for university\\3
kyrs\\6 семетр\\java\\тесты"));
        if (fc.showOpenDialog(this) ==
javax.swing.JFileChooser.APPROVE_OPTION) {
            try (java.io.ObjectInputStream ois = new java.io.ObjectInputStream(new
java.io.FileInputStream(fc.getSelectedFile()))) {
                Object readData = ois.readObject();
                if (readData instanceof java.util.ArrayList) {
                    dataList.clear();
                    dataList.addAll((java.util.ArrayList<RecIntegral>) readData);
                    fillActionPerformed(evt);
                }
            } catch (Exception ex) {
                javax.swing.JOptionPane.showMessageDialog(this, "Ошибка
десериализации: " + ex.getMessage());
            }
        }
    }

    // Variables declaration - do not modify
    private javax.swing.JButton add;
    private javax.swing.JButton clac;
    private javax.swing.JButton clearBtn;
    private javax.swing.JButton delete;
    private javax.swing.JTextField downpr;
    private javax.swing.JButton fillBtn;
    private javax.swing.JLabel jLabel1;
    private javax.swing.JLabel jLabel2;
    private javax.swing.JLabel jLabel3;
    private javax.swing.JLabel jLabel4;
    private javax.swing.JPanel jPanel1;
    private javax.swing.JPanel jPanel2;
    private javax.swing.JPanel jPanel3;
    private javax.swing.JPanel jPanel4;
    private javax.swing.JScrollPane jScrollPane4;
    private javax.swing.JTable jTable1;
    private javax.swing.JButton loadBinBtn;
    private javax.swing.JButton loadTxtBtn;
```

```
    private javax.swing.JButton saveBinBtn;
    private javax.swing.JButton saveTxtBtn;
    private javax.swing.JTextField st3p;
    private javax.swing.JTextField uppr;
    // End of variables declaration

    public static void main(String[] args) {
        java.awt.EventQueue.invokeLater(() -> {
            new MainFrame().setVisible(true);
        });
    }
}
```

Ожидаемые изменения:
Приложение должно работать хорошо независимо от выполнения и нагрузки расчета. Не должно быть осложнений в добавлении и использования программы во время расчета тяжелых значений.

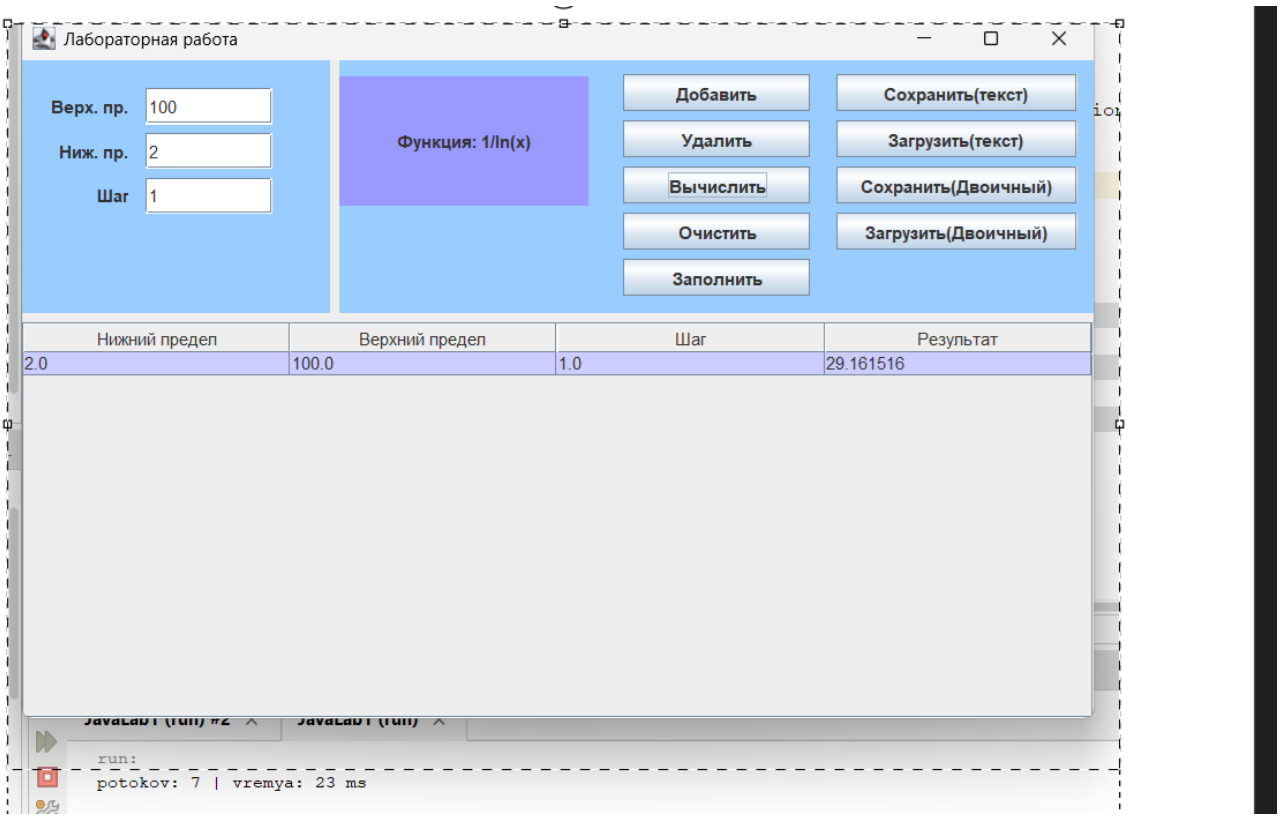
Проверка изменений:





проверим с одни потоком. Видна задержка в 18 ms

Проверим так же с большим количеством потоков



проверка с 7 потоками. Задержка в 23 ms.

Взяли тяжелые значения. Добавляли в таблицу и вычисляли независимо от того, что идет расчет. В зависимости от данных расчет может происходить быстрее для определенных данных. Провели разницу в многопоточности и однопоточности.

Вывод: научились создавать многопоточные приложения с использованием стандартных средств языка Java. Модифицировали программу для оптимизации возможностей.